



TERRAGUARD
Team

INDRADHANU 2026

PCCOE International Grand Challenge

Theme: AI for Climate Change

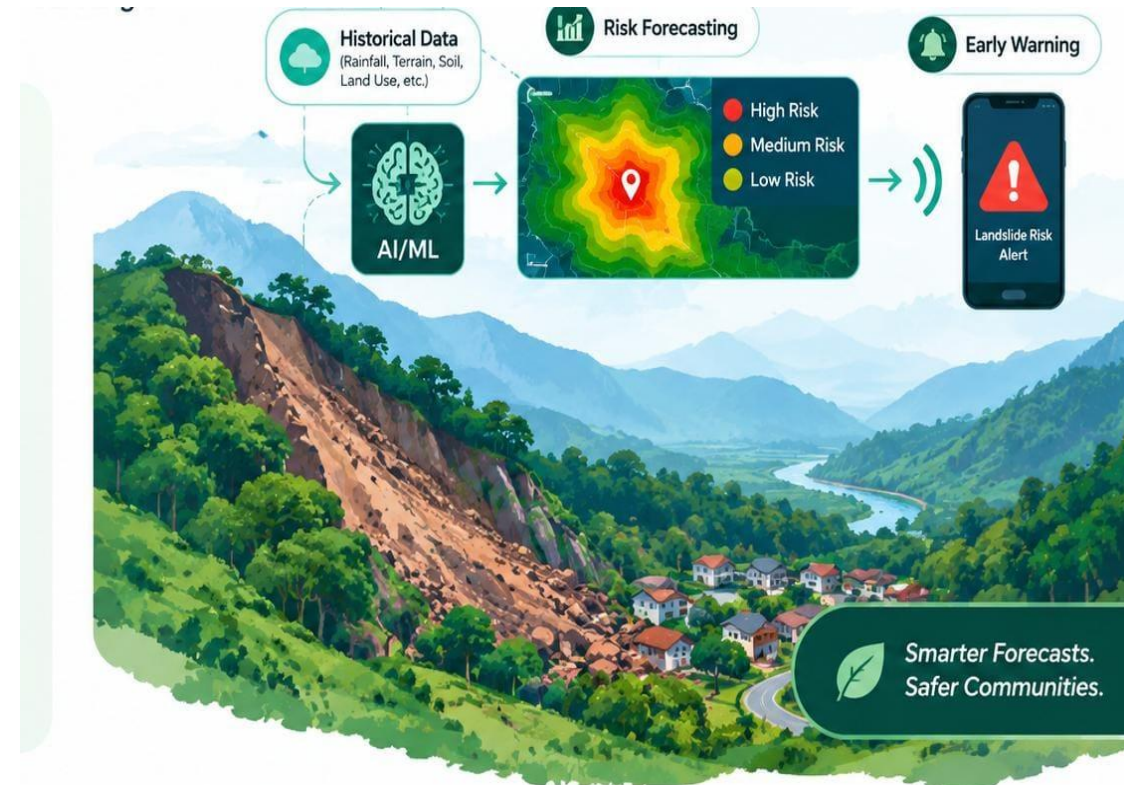
Historical-Data-Based Landslide Risk Prediction Platform

Team Name : Terraguard

Team Members (UG) :

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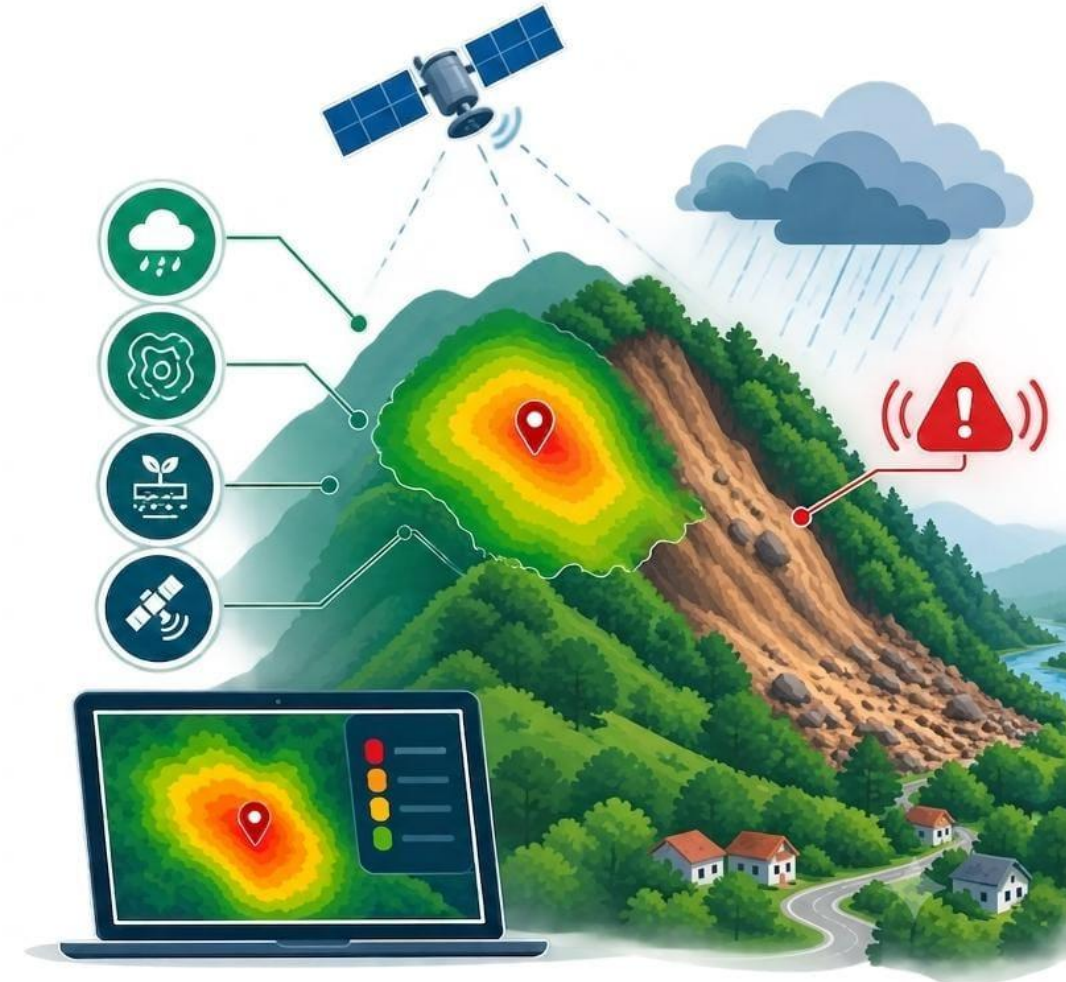
Expected Outcomes

06

Conclusion / Reference

Key Takeaways

AI • Risk Forecasting • Early Warning



Problem Statement

Landslides are a major natural hazard in hilly and mountainous regions, where factors such as heavy rainfall, soil moisture, slope, elevation, geology, and land cover can contribute to slope instability. Existing methods often rely on static susceptibility maps or individual environmental factors, which may not adequately show how landslide risk changes over time.

- ❖ **Rising Climate Risk** — Erratic rainfall from climate change is triggering more frequent landslides in hilly terrain.
- ❖ **Unused Historical Data** — Rainfall, terrain, and soil data already exist across agencies but are never combined for prediction.
- ❖ **Slow, Generic Warnings** — Current alerts are broad and delayed, leaving little time for at-risk communities to act.
- ❖ **High Exposure, No Tools** — Remote hill regions face the highest risk with the least access to predictive monitoring.
- ❖ **No Software-Only Solution** — Most proposed systems rely on costly sensor hardware instead of existing open data.





Objectives

 **Bring Data Together:** Combine rainfall, soil, terrain, and past landslide data in one place.

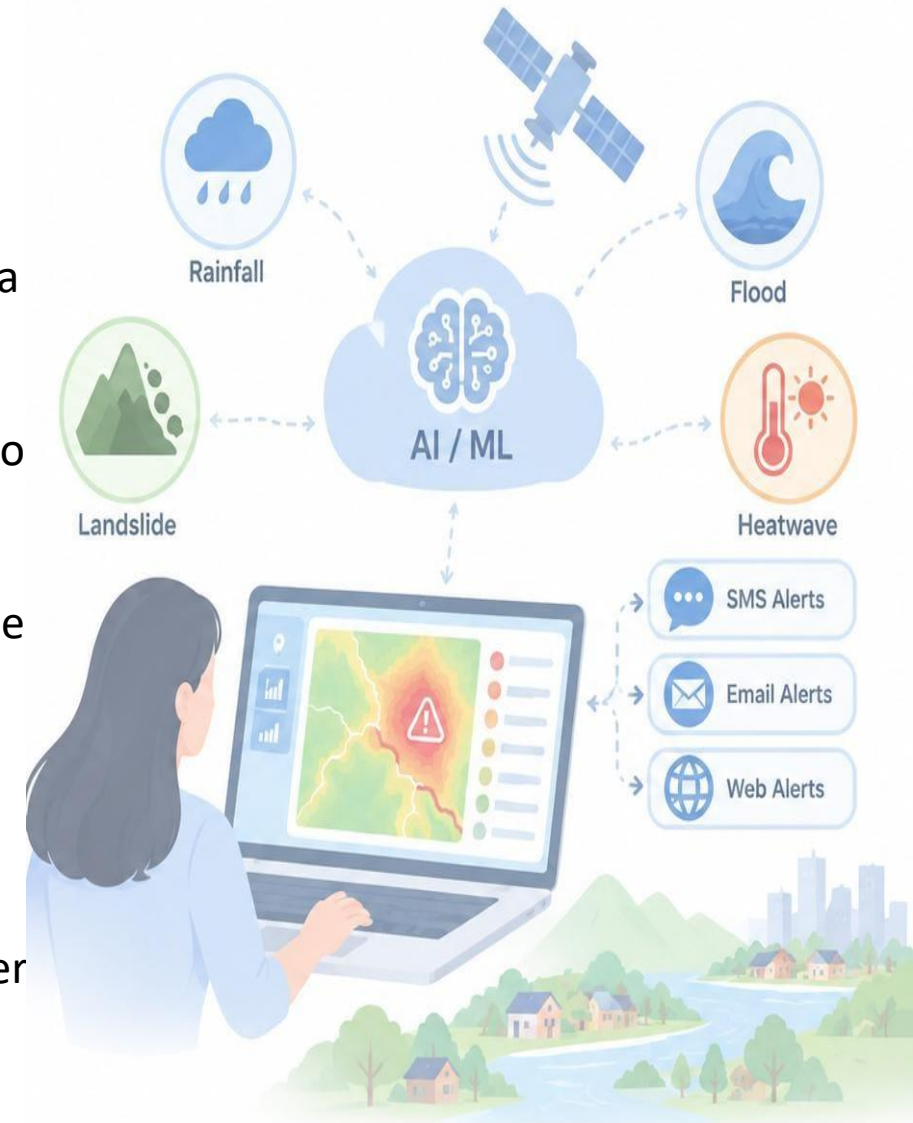
 **Predict Risk Early:** Use AI to identify areas that may become dangerous before a landslide occurs.

 **Send Timely Warnings:** Alert people through SMS, email, or web notifications so they can act early.

 **Make Risk Easy to Understand:** Show danger zones and safer routes on a simple map.

 **No Hardware Needed:** Work entirely with existing data, without installing sensors or equipment.

 **Ready for the Future:** Easily expand the system to floods, cloudbursts, and other regions.





OUR SOLUTION



Historical Data Aggregation

Consolidates years of rainfall, soil moisture, terrain (DEM/slope) and past landslide records from open archives.



AI/ML Risk Modelling

Models trained on this historical data classify each region's risk as Low, Moderate, High or Critical.



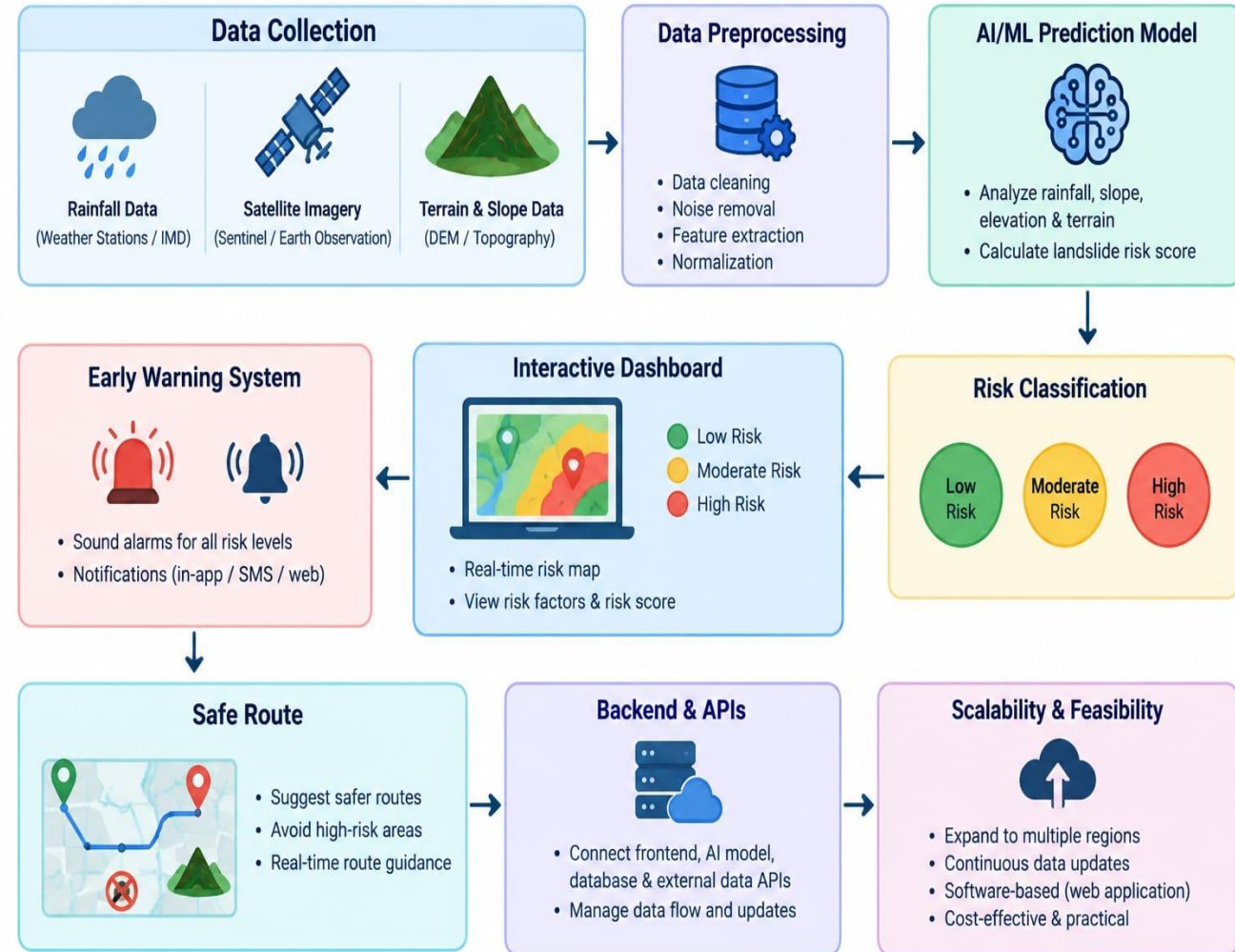
Automated Early Warnings

Instant SMS, email and web alerts to authorities and communities the moment risk crosses a threshold.



Interactive Risk Dashboard

Web-based GIS map of risk zones, trends and safe routes — accessible anywhere, no hardware required.



Our Goal: Build a reliable, scalable, software-first landslide forecasting system that saves lives using data already available today.



TECHNOLOGY STACK

Frontend



React.js, HTML5, CSS3, JavaScript

Backend



Python — FastAPI / Flask, REST APIs

AI / ML



Python, Pandas, NumPy, Scikit-learn, XGBoost

Database



PostgreSQL + PostGIS (spatial queries)

GIS & Mapping



Leaflet.js / Mapbox interactive risk maps

Alerts & Notifications



Twilio SMS API, SMTP Email, Web Push

Historical Data Sources



IMD rainfall, NASA SMAP, ISRO Bhuvan, GSI, Sentinel-2

Cloud Deployment



Fully hosted on cloud (AWS / Render) — zero hardware

HOW IT WORKS — PROCESS FLOW



Historical Data Collection

Rainfall, soil moisture, terrain and past landslide records from open archives



Preprocessing & Feature Engineering

Cleaning, alignment and derivation of rainfall, slope and saturation features



AI/ML Model Training

Supervised learning on historical patterns (Random Forest / XGBoost)



Risk Classification

Each zone scored Low / Moderate / High / Critical



GIS Risk Mapping

Risk levels visualised on an interactive web map



Early Warning & Alerts

Automated SMS, email and dashboard notifications

Entirely cloud/web-based for this phase — no on-ground hardware or live sensors required.



 **Spatiotemporal Forecasting** – Predicts where and when risk may increase.

 **Multi-Source Data** – Combines rainfall, soil moisture, terrain, geology, and past landslides.

 **Advanced AI** – Uses CNN + BiLSTM + Attention for better pattern analysis.

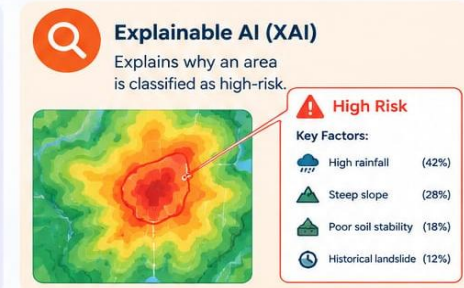
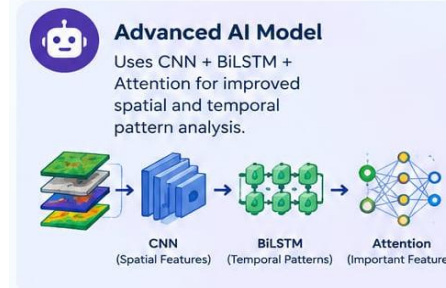
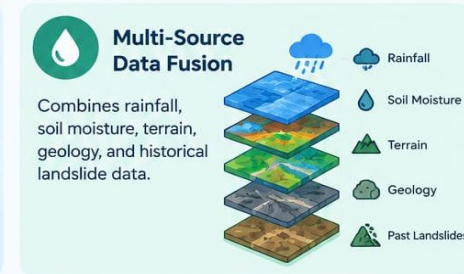
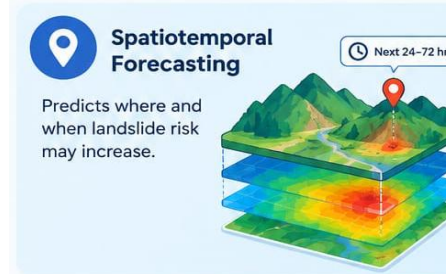
 **Explainable AI** – Shows why an area is risky.

 **Dynamic Risk Map** – Displays High / Medium / Low risk zones.

 **Safe Route Intelligence** – Finds risky areas on a route and suggests safer alternatives.

 **Offline Support** – Provides access to risk information during poor connectivity.

 **Future Ready** – Can support real-time monitoring and alerts in the future.



 **Early Prediction:** Predicts where and when landslide risk may increase.

 **Smart AI Analysis:** Uses AI to analyse complex environmental data accurately.

 **Risk Identification:** Classifies areas into  High,  Medium, and  Low-risk zones.

 **Safer Travel:** Suggests safer routes by identifying risky locations.

 **Early Warning Support:** Can help provide alerts before dangerous situations occur.

 **Offline Access:** Risk information can be available even with poor connectivity.





Historical Rainfall Data

India Meteorological Department (IMD) — Open climate data archives

mausam.imd.gov.in



Historical Soil Moisture

NASA SMAP L4 (Soil Moisture Active Passive), via Google Earth Engine catalog

earthengine.google.com



Terrain & Slope Data

ISRO Bhuvan — DEM and disaster-thematic geospatial layers

bhuvan.nrsc.gov.in



Past Landslide Inventory

Geological Survey of India — Bhukosh geoscience data portal

bhukosh.gsi.gov.in



Satellite Imagery

Copernicus Sentinel-2 (European Space Agency)

dataspace.copernicus.eu



Machine Learning

Scikit-learn & XGBoost documentation

scikit-learn.org



Geospatial Analysis

PostGIS spatial database documentation

postgis.net